IILM UNIVERSITY, GREATER NOIDA

SEMICONDUCTOR AND QUANTUM PHYSICS

Course Code: 26SOST101

COMPLETE ASSIGNMENT ANSWER SHEET

Prepared in the same Unit 1-4 order as the supplied assignment. The answers are exam-oriented and include derivations, formulas and schematic diagram descriptions where required.

UNIT 1 — SEMICONDUCTORS, FERMI-DIRAC STATISTICS & SUPERCONDUCTIVITY

1. Briefly explain metals, semiconductors and insulators based on band gap.

In band theory, the energy difference between the bottom of the conduction band (E_c) and the top of the valence band (E_v) is the forbidden energy gap E_g. **Metals** have overlapping bands or a partially filled band, so E_g is effectively zero and conduction is easy. **Semiconductors** have a small finite E_g, so thermal energy can excite some electrons into the conduction band. **Insulators** have a large E_g, so very few electrons can reach the conduction band at ordinary temperatures.

2. With the help of a diagram define valence band, conduction band and Fermi level.

The **valence band** is the highest normally occupied allowed band. The **conduction band** is the allowed band above it in which electrons can move and conduct current. The **Fermi level E_F** is the energy for which the occupation probability is one-half at finite temperature. In an intrinsic semiconductor it lies approximately near the centre of the forbidden gap.

Schematic: **Conduction band (E_c)** | forbidden gap E_g | **Fermi level E_F** | **Valence band (E_v)**.

3. Differentiate between n-type and p-type extrinsic semiconductors.

n-type: formed by pentavalent donor impurities; electrons are majority carriers, holes are minority carriers, and the donor level is close to the conduction band. **p-type:** formed by trivalent acceptor impurities; holes are majority carriers, electrons are minority carriers, and the acceptor level is close to the valence band.

4. Write the concept of holes. When are the effective mass of holes assumed same as that of electron?

A hole is the vacancy created by removal of an electron from a nearly filled valence band. It behaves as a mobile positive charge carrier of charge +e. The effective masses of electrons and holes are generally different because they depend on band curvature. They may be taken as equal only when the problem explicitly makes that simplifying approximation.

5. Schematically show how F(E), the Fermi function, varies with energy at 0 K and finite temperature.

The Fermi function is f(E)=1/[1+exp((E-E_F)/kT)]. At 0 K it is a sharp step: f=1 below E_F and f=0 above E_F. At finite temperature the step becomes smooth, with f(E_F)=1/2. Thus thermal energy creates a small population of electrons above E_F and vacancies below it.

6. Discuss the characteristics of superconductors in superconducting state.

Main characteristics are: (i) essentially zero dc resistance below T_c; (ii) Meissner effect; (iii) a critical temperature T_c; (iv) a critical magnetic field H_c; (v) a critical current; and (vi) persistent current with negligible voltage drop. The superconducting state is destroyed if temperature, magnetic field or current exceeds its critical value.

7. Write a short note on Meissner effect.

The Meissner effect is the expulsion of magnetic flux from the bulk of a material when it is cooled below its critical temperature in a magnetic field. A superconductor therefore behaves as a strong diamagnet in the superconducting state. This is a defining property of superconductivity and not merely a consequence of zero resistance.

8. With the help of an energy diagram explain formation of bands in solids. Differentiate semiconductor, conductor and insulator.

An isolated atom has discrete energy levels. In a crystal, a very large number of atoms are brought close together. Interatomic interaction and the Pauli exclusion principle split each atomic level into a large number of closely spaced levels, forming allowed energy bands. Between allowed bands are forbidden energy gaps.

The valence band is the highest occupied band and the conduction band is the next available band. The band-theory

classification is: conductor - overlapping/partially filled bands; semiconductor - small forbidden gap; insulator - large forbidden gap.

9. What is Fermi-Dirac distribution function? Give its physical significance and temperature dependence.

The Fermi-Dirac distribution gives the probability that an available state of energy E is occupied by an electron: **f(E)=1/[1+exp((E-E_F)/kT)]**. It is required because electrons are fermions and obey the Pauli exclusion principle. At 0 K, all states below E_F are occupied and those above E_F are empty. At finite temperature the transition becomes smooth and f(E_F)=1/2.

10. What is superconductivity? Discuss temperature dependence of resistivity and the significance of T_c and H_c.

Superconductivity is the state in which certain materials exhibit essentially zero electrical resistance and magnetic flux expulsion below a critical temperature. As a superconducting material is cooled, its normal-state resistivity decreases; at T_c it falls abruptly to essentially zero. **T_c** is the maximum temperature at which superconductivity persists. **H_c** is the critical magnetic field above which superconductivity is destroyed. In general, H_c decreases as T approaches T_c.

11. Deduce carrier concentration in intrinsic semiconductors and prove the intrinsic Fermi level lies near the middle of the forbidden band.

For a non-degenerate semiconductor, electron concentration is n=N_c exp[-(E_c-E_F)/kT] and hole concentration is p=N_v exp[-(E_F-E_v)/kT]. For an intrinsic semiconductor n=p=n_i. Equating the two gives **E_Fi=(E_c+E_v)/2 +(kT/2)ln(N_v/N_c)**. Since N_c and N_v are proportional to the effective density of states, the correction term is usually small. Under the common approximation N_c=N_v, **E_Fi=(E_c+E_v)/2**, exactly at the centre of the gap. Also, **n_i=sqrt(N_c N_v) exp[-E_g/(2kT)]**.

UNIT 2 — P-N JUNCTIONS, BANDGAPS & PHOTOVOLTAIC CELLS

1. Differentiate between direct and indirect bandgap.

In a **direct-gap** semiconductor, the conduction-band minimum and valence-band maximum occur at the same crystal momentum. Optical recombination can occur without a phonon for momentum conservation, so light emission is efficient. In an **indirect-gap** semiconductor they occur at different momenta; a phonon is needed, making radiative emission less efficient.

2. Write the expression of efficiency of photovoltaic cell.

Photovoltaic conversion efficiency is **eta=P_max/P_in x 100%**, where P_max=V_mp I_mp. Hence eta=(V_mp I_mp/P_in)x100%. Using fill factor, eta=(V_oc I_sc FF/P_in)x100%.

3. Explain biasing in a p-n junction and its two types.

Biasing means applying an external voltage across a p-n junction. In **forward bias**, p is connected to positive and n to negative; the barrier and depletion width decrease and current rises strongly. In **reverse bias**, p is negative and n is positive; the barrier and depletion width increase and only a small reverse current flows until breakdown.

4. Briefly describe the depletion region.

When a p-n junction is formed, electrons diffuse from n to p and holes from p to n and recombine. Immobile ionized donor and acceptor ions remain near the junction. This carrier-depleted region is the depletion region. The uncovered charges create an internal electric field and built-in potential that oppose further diffusion.

5. Discuss photovoltaic effect.

The photovoltaic effect is the generation of voltage and current when light is absorbed by a material. In a p-n junction, photons create electron-hole pairs. The built-in electric field separates the carriers and the contacts collect them, producing a photovoltage and photocurrent through an external load.

6. What is a p-n junction diode? Explain formation of barrier potential and depletion region.

A p-n junction diode is a semiconductor device formed by joining p-type and n-type regions. Majority carriers initially diffuse across the junction because of the concentration gradient. Recombination leaves negative acceptor ions on the p-side and positive donor ions on the n-side. These fixed charges form the depletion region. Their electric field produces the built-in barrier potential, which opposes further majority-carrier diffusion. At equilibrium, drift and diffusion currents balance.

7. Explain carrier flow in forward and reverse bias and draw/describe I-V characteristics.

In forward bias the external field opposes the junction field, reducing the barrier and depletion width. Majority carriers cross the junction and current increases rapidly after the knee region. In reverse bias the barrier increases; a small reverse saturation current flows due mainly to minority carriers. At sufficiently large reverse voltage, breakdown causes a sharp current increase. The diode equation is **I=I_s[exp(qV/(eta kT))-1]**.

8. What is a photovoltaic cell? Explain its working with a suitable diagram.

A photovoltaic cell is a semiconductor p-n junction device that converts light into electrical energy. A typical structure contains a front contact, semiconductor junction/absorber, rear contact and protective/anti-reflection layer. Light creates electron-hole pairs. The junction field separates them: electrons move toward the n-side and holes toward the p-side. The contacts collect the carriers and current flows through an external load.

Schematic: **Sunlight -> front contact -> p-n junction -> rear contact -> external load**.

9. Compare direct and indirect bandgap semiconductors using suitable energy-band diagrams.

Direct gap: E_c minimum and E_v maximum occur at the same k; photon emission is comparatively efficient. Indirect gap: they occur at different k; a phonon participates in the transition. Direct-gap materials are therefore preferred for

efficient LEDs and semiconductor lasers, whereas indirect-gap materials are widely used for electronic devices and solar applications.

UNIT 3 — QUANTUM PHYSICS

1. State de Broglie's hypothesis and determine the wavelength of a particle.

de Broglie proposed that every moving material particle has an associated matter wave. Its wavelength is **lambda=h/p**. For p=mv, lambda=h/(mv). Since non-relativistic kinetic energy E=p^2/(2m), p=sqrt(2mE), hence **lambda=h/sqrt(2mE)**.

2. State inadequacies of classical physics.

Classical physics could not adequately explain black-body radiation, the photoelectric effect, atomic line spectra, atomic stability and the wave nature of microscopic particles. These failures led to quantum ideas such as energy quantization, photons and matter waves.

3. Define wave function psi and state its physical significance.

The wave function psi describes the quantum state of a particle. It is not itself directly observable. According to Born's interpretation, **|psi|^2** is the probability density of finding the particle at a specified position and time. A physically acceptable wave function must be finite, single-valued and continuous under the usual conditions.

4. Briefly discuss normalization condition.

The total probability of finding a particle somewhere in all space must equal one: **integral |psi|^2 d tau = 1**. If a trial wave function does not satisfy this condition, it is multiplied by a normalization constant.

5. State quantum tunnelling effect.

Quantum tunnelling is the transmission of a particle through a potential barrier even when its kinetic energy is less than the barrier height. The wave function penetrates the classically forbidden region and decays there, leaving a finite transmission probability.

6. Briefly explain quantum computing.

Quantum computing uses qubits. Unlike classical bits, a qubit can exist in a superposition of basis states, and qubits can be entangled. Quantum gates manipulate amplitudes and measurement produces probabilistic outcomes. Quantum computers are promising for selected tasks such as quantum simulation and some cryptographic/search problems.

7. Derive the time-dependent Schrodinger wave equation.

For a plane matter wave, **psi=A exp[i(kx-omega t)]**. Using p=hbar k and E=hbar omega, differentiation gives i hbar (d psi/dt)=E psi and -hbar^2/(2m)(d^2 psi/dx^2)=(p^2/2m)psi. For a particle with potential V, E=p^2/(2m)+V. Therefore, **i hbar dpsi/dt=[-hbar^2/(2m) d^2/dx^2 + V]psi**. In three dimensions, d^2/dx^2 is replaced by the Laplacian nabla^2.

8. Derive the time-independent Schrodinger equation and explain physical significance of psi.

For a time-independent potential, write psi(x,t)=phi(x)exp(-iEt/hbar). Substitution into the time-dependent equation gives **-(hbar^2/2m)d^2phi/dx^2 + V phi = E phi**. Equivalently, d^2phi/dx^2 +(2m/hbar^2)(E-V)phi=0. The wave function determines probabilities through |phi|^2. Allowed stationary solutions satisfy boundary conditions and have definite energy eigenvalues.

9. Apply the time-independent equation to a particle in a one-dimensional box.

For an infinite well from x=0 to L, V=0 inside and psi(0)=psi(L)=0. The equation becomes d^2psi/dx^2+k^2psi=0, where k^2=2mE/hbar^2. The boundary conditions give kL=n pi, n=1,2,3,... . Thus **E_n=n^2 h^2/(8mL^2)**. Normalization gives **psi_n=sqrt(2/L) sin(n pi x/L)**, for 0<x<L. Outside the box the wave function is zero.

10. Compare classical and quantum predictions for a particle encountering a potential barrier.

Classically, if E<V_0 the particle cannot cross the barrier. Quantum mechanically, the wave function penetrates the barrier and a finite transmission probability exists: this is tunnelling. For a rectangular barrier with E<V_0, the wave

function inside varies approximately as exp(-kappa x), where **kappa=sqrt[2m(V_0-E)]/hbar**. Transmission therefore decreases rapidly as barrier width or barrier height increases.

11. What are matter waves? Show lambda=h/sqrt(2mE).

Matter waves are waves associated with moving particles. de Broglie's relation gives lambda=h/p. For a non-relativistic particle, kinetic energy E=p^2/(2m), hence p=sqrt(2mE). Substitution gives **lambda=h/sqrt(2mE)**.

UNIT 4 — LASERS AND OPTICAL FIBRES

1. Discuss necessary conditions for lasing action.

The necessary conditions are a suitable active medium, an energy-pumping mechanism, population inversion, stimulated emission and optical feedback. The population of the upper laser level must be sufficiently large so that stimulated emission exceeds absorption. A resonator keeps photons interacting repeatedly with the gain medium.

2. Identify the basic components of a laser and explain the function of each.

Active medium: provides the gain transition. **Pumping source:** supplies energy and establishes population inversion. **Optical resonator:** provides feedback and mode selection. **Output coupler:** a partially transmitting mirror that lets a fraction of the amplified radiation emerge as the laser beam.

3. Explain importance of an optical resonator.

The optical resonator reflects light repeatedly through the gain medium, increasing amplification. It selects supported optical modes, establishes directional output and allows stimulated emission to build up until laser threshold is reached.

4. What is the working principle of an optical fibre? Name and define it.

An optical fibre works on **total internal reflection**. It consists mainly of a high-index core surrounded by lower-index cladding. Light entering within the acceptance cone strikes the core-cladding boundary at an angle greater than the critical angle and is repeatedly reflected along the fibre. The fibre's light-guiding principle is therefore TIR.

5. Briefly describe the working principle of a semiconductor laser.

A semiconductor laser is usually a forward-biased p-n junction with a thin active region. Carrier injection creates a high carrier concentration. Electron-hole recombination produces photons, and stimulated emission is enhanced by optical feedback from reflecting facets or mirrors. Above threshold current, coherent laser output is obtained.

1. Derive relations between Einstein A and B coefficients using thermal equilibrium and Planck's law.

Consider two levels E_1 and E_2 with E_2-E_1=h nu and populations N_1,N_2. Absorption rate = N_1 B_12 rho(nu). Spontaneous emission rate = N_2 A_21. Stimulated emission rate = N_2 B_21 rho(nu). At thermal equilibrium, upward rate equals total downward rate:

N_1 B_12 rho = N_2 A_21 + N_2 B_21 rho.

Thus rho = N_2 A_21/(N_1 B_12-N_2 B_21). Using the Boltzmann population ratio N_2/N_1=(g_2/g_1)exp(-h nu/kT) and comparing with Planck's radiation law yields **g_1 B_12=g_2 B_21** and **A_21/B_21=8 pi h nu^3/c^3** for the standard spectral-energy-density convention. For equal degeneracies, B_12=B_21.

2. Explain construction and working of a He-Ne laser with energy-level scheme and applications.

A He-Ne laser contains a low-pressure helium-neon gas mixture in a discharge tube. Electrodes produce an electrical discharge and mirrors form an optical resonator; one mirror is partially transmitting. The discharge excites helium atoms to metastable states. Collisions transfer this energy to nearby neon excited states, producing population inversion in neon. Stimulated emission is amplified in the cavity and output emerges through the partially transmitting mirror. A common He-Ne wavelength is about 632.8 nm (red).

Energy scheme: electrical pumping -> excited He metastable states -> resonant energy transfer to Ne -> upper Ne laser state -> stimulated emission -> lower Ne state -> further decay. **Applications:** alignment, surveying, interferometry, holography, laboratory optics and precision measurement.

3. Explain construction and working of Ruby laser and describe its energy levels.

Ruby laser is a three-level solid-state laser. The active medium is a ruby crystal, Al2O3 doped with Cr3+ ions. A flash lamp optically pumps the chromium ions and the rod is placed in a resonator with one partially transmitting end. Pumping excites ions to higher absorption bands; they quickly decay non-radiatively to a metastable level. Population accumulates there and stimulated emission from the metastable level to the ground state produces red laser light, approximately 694.3 nm. Strong pumping is needed because it is a three-level system.

4. Discuss classification of optical fibres. Explain step-index and graded-index; distinguish single-mode and multimode.

Fibres are classified by refractive-index profile and number of modes. **Step-index:** core has uniform n_1 and cladding has lower uniform n_2; the index changes abruptly at the boundary. **Graded-index:** core index decreases gradually from centre to edge, so rays follow curved paths and modal dispersion is reduced. **Single-mode:** only the fundamental mode propagates, normally when V<2.405; it has very low modal dispersion and is suitable for long-distance links. **Multimode:** many modes propagate, giving greater modal dispersion but easier coupling and common use over shorter distances.

5. How do acceptance angle and numerical aperture control entry and propagation of light? Derive expressions.

The **acceptance angle theta_a** is the maximum external half-angle at which light can enter and still be guided. The **numerical aperture NA** measures the light-gathering ability. For a step-index fibre with core index n_1, cladding index n_2 and external medium index n_0, Snell's law and the critical-angle condition give **n_0 sin(theta_a)=sqrt(n_1^2-n_2^2)**. Therefore **NA=n_0 sin(theta_a)=sqrt(n_1^2-n_2^2)**. For air n_0 approximately 1, so NA=sin(theta_a). Larger NA gives a larger acceptance cone and easier coupling of light.

FORMULA & DIAGRAM REVISION SHEET

Key formulae

Topic	Formula	Use
Fermi-Dirac	f(E)=1/[1+exp((E-E_F)/kT)]	Occupation probability
Intrinsic semiconductor	n_i=sqrt(N_c N_v) exp[-E_g/(2kT)]	Intrinsic carrier concentration
Diode	I=I_s[exp(qV/(eta kT))-1]	p-n junction I-V
de Broglie	lambda=h/p=h/sqrt(2mE)	Matter waves
1-D box	E_n=n^2h^2/(8mL^2)	Quantized energy
1-D box wavefunction	psi_n=sqrt(2/L) sin(n pi x/L)	Normalized state
Einstein	g_1B_12=g_2B_21; A_21/B_21=8 pi h nu^3/c^3	Laser theory
Fibre NA	NA=sqrt(n_1^2-n_2^2)=n_0 sin(theta_a)	Light acceptance
Single-mode	V=(2 pi a/lambda)NA < 2.405	Single-mode condition

Schematic diagrams to reproduce in an exam

Band diagram: Conduction band (E_c) | Forbidden gap E_g | Fermi level E_F | Valence band (E_v)

p-n junction: p-side | depletion region + barrier potential | n-side

Laser: Fully reflecting mirror | Active/gain medium | Partially transmitting mirror -> output

Fibre: Lower-index cladding / Higher-index core: zig-zag guided ray / Lower-index cladding

End of complete answer sheet.